Grouped frequency tables and histograms

1

a Ungrouped

b Grouped

2

a 33.5

b 21

3 $19 < \text{height} \leq 28$

4

a 7

b 4

5

a

Score (x)	Frequency (f)
$10 - 19$	3
$20 - 29$	4
$30 - 39$	2
$40 - 49$	1
$50 - 59$	1

b

Score (x)	Frequency (f)
$30 - 39$	5
$40 - 49$	4
$50 - 59$	2
$60 - 69$	8
$70 - 79$	3
$80 - 89$	3

6

a $20 \rightarrow 29, 30 \rightarrow 39, 40 \rightarrow 49, 50 \rightarrow 59,$ and $60 \rightarrow 69$

b

Score (x)	$20 \rightarrow 29$	$30 \rightarrow 39$	$40 \rightarrow 49$	$50 \rightarrow 59$	$60 \rightarrow 69$
Frequency (f)	3	2	1	3	2

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7

a $10 \to 19, 20 \to 29, 30 \to 39, 40 \to 49, 50 \to 59, 60 \to 69,$ and $70 \to 79$

b

Score (x)	Frequency (f)
$10 \to 19$	7
$20 \to 29$	3
$30 \to 39$	2
$40 \to 49$	3
$50 \to 59$	6
$60 \to 69$	4
$70 \to 79$	4

8

a

Score (x)	Frequency (f)
$1 - 5$	6
$6 - 10$	12
$11 - 15$	7
$16 - 20$	3
$21 - 25$	9
$26 - 30$	2
$31 - 35$	4

b

Score (x)	Frequency (f)
$20 - 29$	150
$30 - 39$	140
$40 - 49$	160
$50 - 59$	150
$60 - 69$	130
$70 - 79$	110
$80 - 89$	120
$90 - 99$	130

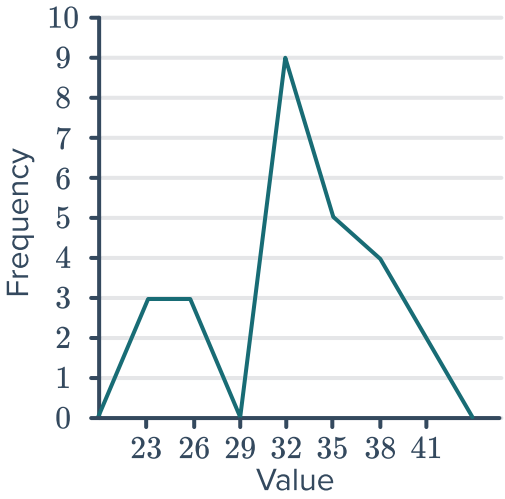
9

a

Class interval	Class centre	Frequency
$21.5 \leq p < 24.5$	23	3
$24.5 \leq p < 27.5$	26	3
$27.5 \leq p < 30.5$	29	0
$30.5 \leq p < 33.5$	32	9
$33.5 \leq p < 36.5$	35	5
$36.5 \leq p < 39.5$	38	4
$39.5 \leq p < 42.5$	41	2

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b



c 0

d 17

e 15

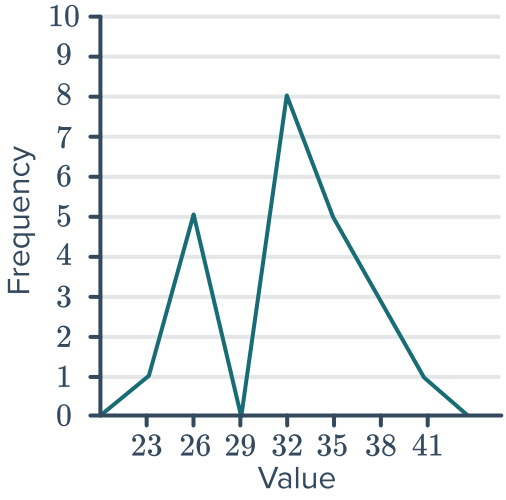
f 20

10

a

Class limits	Midpoint	Frequency
$21.5 - 24.5$	23	1
$24.5 - 27.5$	26	5
$27.5 - 30.5$	29	0
$30.5 - 33.5$	32	8
$33.5 - 36.5$	35	5
$36.5 - 39.5$	38	3
$39.5 - 42.5$	41	1

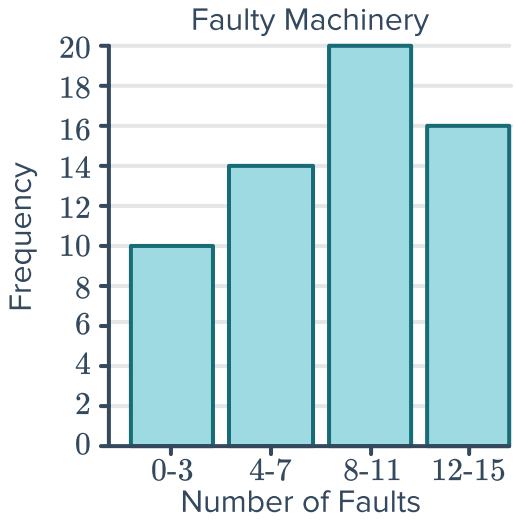
b



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14

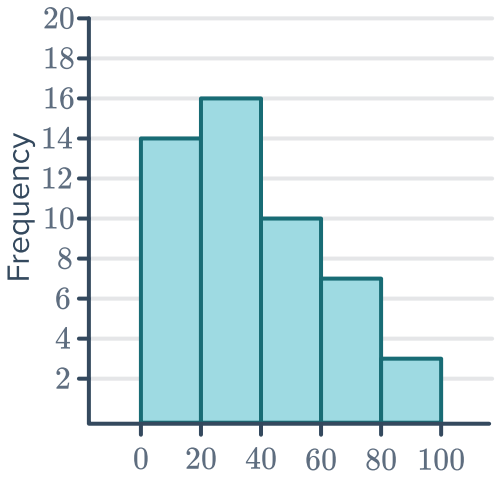
a



b 0

15

a



b

i No **ii** Yes **iii** No **iv** No

16

a

i $0 \leq \text{rate} < 5$

ii $5 \leq \text{rate} < 10$

iii $10 \leq \text{rate} < 15$

iv $15 \leq \text{rate} < 20$

b 3

17

a 140.9 cents

b 11 days

18

a 25 students **b** 13 students **c** 16%

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19

a

Height (cm)	Frequency
$95 \leq h < 100$	1
$100 \leq h < 105$	3
$105 \leq h < 110$	2
$110 \leq h < 115$	12
$115 \leq h < 120$	19
$120 \leq h < 125$	15
$125 \leq h < 130$	4
$130 \leq h < 135$	4
Total:	60

b 31.7%

20

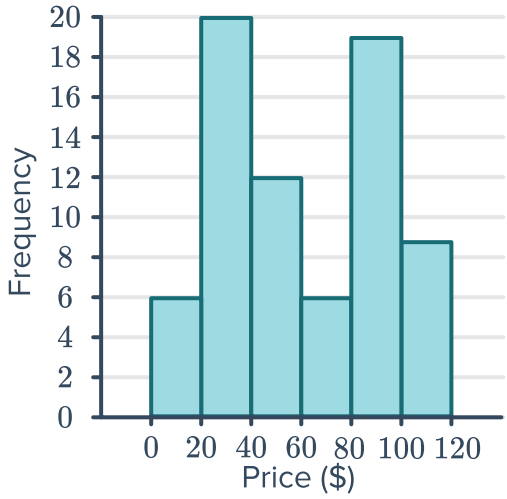
a

Speed (km/h)	Frequency
$110 \leq s < 120$	4
$120 \leq s < 130$	12
$130 \leq s < 140$	8
$140 \leq s < 150$	5
$150 \leq s < 160$	2
Total:	31

b 51.6%

21

a



b $20 - 39$ and $80 - 99$

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c The data shows that there are two main price ranges of books that were recently bought by university students: $\$20$ to $\$39$, and $\$80$ to $\$99$.

22

a

Battery life (minutes)	Frequency
$400 \leq x < 500$	3
$500 \leq x < 600$	9
$600 \leq x < 700$	13
$700 \leq x < 800$	32
$800 \leq x < 900$	22
$900 \leq x < 1000$	27
$1000 \leq x < 1100$	6
$1100 \leq x < 1200$	4
$1200 \leq x < 1300$	1
Total :	117

b 89.7%

c 9.4%

23

a 35

b $60 \text{ mm} \leq x < 80 \text{ mm}$

c Yes

d No. The table shows ranges of lengths and not individual leaves. We do not know if any of the leaves in the range $0 \leq x < 20$ are less than 5 mm.

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